CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANDUBHAI S. PATEL INSTITUTE OF TECHNOLOGY
U. & P. U. Patel Department of Computer Engineering

Subject Name	:	Data Structures & Algorithms	**Class**	:	B.Tech. (IV Sem)
Subject Code	:	CE245	**Academic year**	:	2019-2020

Practical List

Sr. No.	Aim of the Practical	Hours
1.	Implement Linear Search and Binary Search using array data structure.	2
2.	Implement Sorting Algorithm(s). (a) Bubble Sort (b) Selection Sort (c) Insertion Sort (d) Quick Sort	4
3.	Implement basic operations of stack and queue data structure using an array.	2
4	Implement a program that enters Infix expression, verify validity and convert it into Postfix expression (if valid) .	2
5.	Implement basic operations of circular queue using an array.	2
6.	Implement following operations of singly linked list. (a) Insert a node at front (b) Insert a node at end (c) Insert a node after given node information (d) Delete a node at front (e) Delete a node at last	4
7.	Implement following operations of doubly linked list. (a) Insert a node at front (b) Insert a node at end (c) Insert a node after given node information (d) Delete a node at front (e) Delete a node at last	2

8.	Write a program to construct Binary Search Tree of your mobile phone number and apply different tree traversal techniques on BST. Also search particular digit in BST.	4
9.	Implement a program to implement BFS and DFS Graph traversal techniques.	4
10.	In an array of 20 elements, arrange 15 different values, which are generated randomly between 1,00,000 to 9,99,999. Use hash function to generate key and linear probing to avoid collision. H(x) = (x mod 18) + 2. Write a program to input and display the final values of array.	2

Note:

1. Students have to register in HackerRank platform during first lab.
2. Students are instructed to solve challenges given at the end of lab session.
3. **Evaluation of practical would be based on score of the HackerRank and performance in regular lab sessions.**(grade)